CPM and PERT

Polished Digital Note

Start → Plan and Schedule → Critical Path → Finish

A complete, typed, corrected and reorganized edition

Based on the 31-page handwritten note by **2223036**

Project Scheduling • CPM • PERT • Float • Probability

Prepared as a clean study-ready PDF edition

Editorial Note

This edition converts the handwritten source into a structured digital note. Spelling, grammar, notation, diagrams and arithmetic have been standardized. Where a handwritten value or line was unclear, the result was recomputed from the activity data using standard CPM and PERT rules. The worked examples retain the organization and intent of the source while presenting the calculations in a consistent form.

Notation used throughout

ES	Earliest Start	**EF**	Earliest Finish
LS	Latest Start	**LF**	Latest Finish
TF	Total Float	**FF**	Free Float
IF	Independent Float	t **or** d	Activity duration
a	Optimistic time	m	Most likely time
b	Pessimistic time	t_e	Expected activity time

Contents

1 Project Scheduling

Definition

Project scheduling is the process of arranging project activities in the correct sequence, assigning their start and finish times, estimating resources and costs, and determining the total project completion time.

A good schedule answers four basic questions:

- What work will be done?
- When will it be done?
- Who will do it?
- How long will it take?

1.1 Main Steps of Project Scheduling

1. Identify precedence relationships among activities.
2. Arrange all activities in the correct logical order.
3. Determine activity times and costs.
4. Estimate material, equipment and worker requirements.
5. Determine the critical activities - the activities that cannot be delayed without delaying the project.

1.2 Purpose of Project Scheduling

- Shows the relationships among activities and how they connect.
- Identifies precedence relationships.
- Encourages realistic time and cost estimates.
- Supports better use of resources.
- Provides a basis for planning, monitoring and control.

1.3 Requirements of a Good Scheduling Technique

A good scheduling method should ensure that:

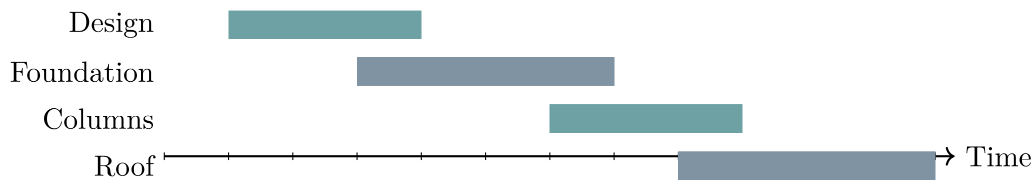
1. every required activity is included;
2. the correct order and precedence are maintained;
3. a duration is assigned to each activity; and
4. total project duration can be calculated.

2 Project Management Scheduling Techniques

The three common techniques in the source are Gantt charts, CPM and PERT.

2.1 Gantt Chart

A Gantt chart is a time-scaled bar chart that shows project activities against calendar time.



Advantages

- Easy to understand and prepare.
- Makes project progress visible.
- Suitable for small and medium projects.

Disadvantages

- Activity dependencies are not always shown clearly.
- Difficult to manage for very large projects.
- The critical path is not obvious.

2.2 Critical Path Method (CPM)

CPM

CPM is a network scheduling technique that identifies the longest sequence of dependent activities. This longest path determines the minimum possible project completion time.

Main uses

- Find project duration.
- Identify critical activities.
- Calculate float or slack.
- Support planning, scheduling, monitoring and control.

CPM is most suitable when activity durations are known or can be treated as fixed.

2.3 Programme Evaluation and Review Technique (PERT)

PERT

PERT is a network scheduling technique used when activity durations are uncertain. It uses three time estimates - optimistic, most likely and pessimistic - to calculate an expected duration.

PERT is especially suitable for research, development and other one-time projects where activity times are uncertain.

3 CPM and PERT: Overview

Both CPM and PERT are used to:

- plan a project;
- schedule activities;
- monitor progress;
- control project completion time; and
- show activity sequence and dependency.

3.1 Historical Background

Technique	Developed by	Year	Early application
CPM	DuPont and Remington Rand	1957	Chemical-plant maintenance and construction planning
PERT	U.S. Navy with Booz Allen Hamilton	1958	Polaris missile programme

3.2 Similarities

Both techniques:

- use a network diagram;
- show precedence relationships;
- identify a critical path; and
- help with project planning, scheduling, monitoring and control.

3.3 Six General Steps

1. Define the project and prepare the Work Breakdown Structure (WBS).
2. Develop activity relationships.
3. Draw the network diagram.
4. Assign activity time and, where relevant, cost.
5. Compute the critical path, i.e., the longest path through the network.
6. Use the network to plan, schedule, monitor and control the project.

Core idea

The critical path is the longest-duration path through the project network. Therefore, the project cannot finish earlier than the time required by that path.

4 CPM Terminology and Calculations

4.1 Forward Pass

The forward pass moves from the start of the network to the finish. It calculates earliest times.

Forward-pass formulas

For activity i with duration d_i:

$$ES_i = \max(EF \text{ of all immediate predecessors})$$

$$EF_i = ES_i + d_i$$

For an activity with no predecessor, $ES = 0$.

4.2 Backward Pass

The backward pass moves from the finish of the network to the start. It calculates latest allowable times without delaying the project.

Backward-pass formulas

For activity i:

$$LF_i = \min(LS \text{ of all immediate successors})$$

$$LS_i = LF_i - d_i$$

For a terminal activity, LF is set equal to the project duration.

4.3 Definitions of ES, EF, LS and LF

ES Earliest possible time an activity can begin.

EF Earliest possible time an activity can finish; $EF = ES + d$.

LS Latest time an activity can start without delaying the project; $LS = LF - d$.

LF Latest time an activity can finish without delaying the project.

4.4 AON Activity-Node Convention

The source uses the following node arrangement for an Activity-on-Node (AON) network:

ES	Activity	EF
LS	Duration	LF
Float / notes		

5 Float, Slack and the Critical Path

Float or slack

Float is the amount of time an activity can be delayed without causing a specified delay elsewhere in the network. The exact meaning depends on the type of float.

5.1 Total Float (TF)

Total float is the amount of time an activity can be delayed without delaying the project completion date.

Total float

$$\text{TF}_i = LS_i - ES_i = LF_i - EF_i$$

For an AOA activity (i, j) with duration t_{ij}:

$$\text{TF}_{ij} = L_j - E_i - t_{ij}$$

Interpretation

- $TF < 0$: the imposed deadline is tighter than the calculated schedule; acceleration, crashing or additional resources may be required.
- $TF = 0$: the activity is critical.
- $TF > 0$: the activity has spare time up to its float limit.

5.2 Free Float (FF)

Free float is the amount of time an activity can be delayed without delaying the earliest start of any immediate successor.

Free float

$$\mathrm{FF}_i = \min(ES \text{ of immediate successors}) - EF_i$$

For an AOA activity (i, j):

$$\mathrm{FF}_{ij} = E_j - E_i - t_{ij}$$

Generally, $FF \leq TF$.

5.3 Independent Float (IF)

Independent float is the delay allowed when predecessors finish as late as permitted and successors still start as early as possible.

Independent float

For an AOA activity (i, j):

$$\mathrm{IF}_{ij} = \max\{0,\ E_j - L_i - t_{ij}\}$$

A convenient AON form is:

$$\mathrm{IF}_i = \max\{0,\ \min ES_{\text{successor}} - \max LF_{\text{predecessor}} - d_i\}$$

5.4 Safety Float (Additional AON Measure)

The handwritten source also records a less commonly used AON measure called safety float. In the source's notation:

$$SF_i = TF_i - (\max LF_{\text{predecessor}} - ES_i),$$

with negative results normally taken as zero.

5.5 Critical Event, Critical Activity and Critical Path

- A **critical event** has zero event slack, i.e., earliest event time equals latest event time.
- A **critical activity** has zero total float.
- The **critical path** is the continuous sequence of critical activities from start to finish. It is the longest-duration path and determines the minimum project completion time.

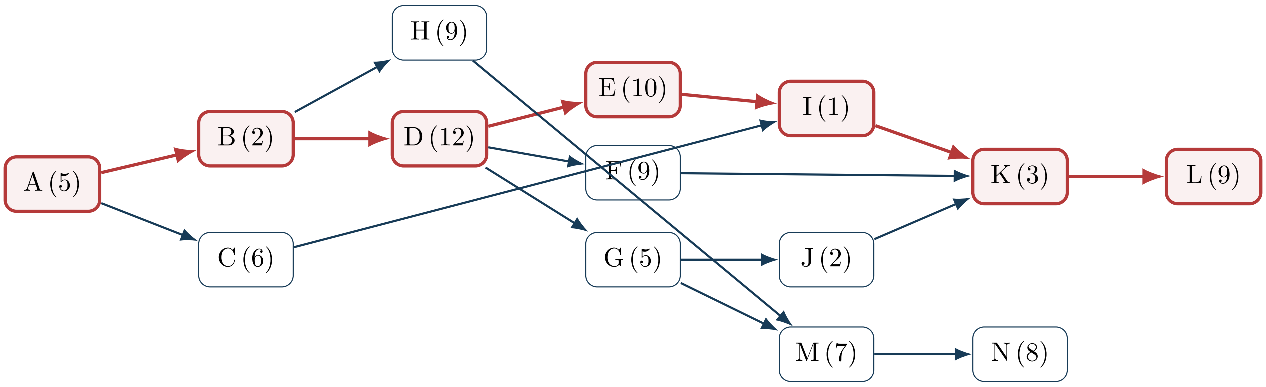
  Any delay in a critical activity delays the whole project unless corrective action is taken.

6 Worked CPM Example 1: AON Network

6.1 Activity Data

Activity	Predecessor(s)	Duration	Activity	Predecessor(s)	Duration
A	–	5	H	B	9
B	A	2	I	C, E	1
C	A	6	J	G	2
D	B	12	K	F, I, J	3
E	D	10	L	K	9
F	D	9	M	H, G	7
G	D	5	N	M	8

6.2 Precedence Network



The critical sequence is highlighted. The critical path is:

A-B-D-E-I-K-L

$$5 + 2 + 12 + 10 + 1 + 3 + 9 = \boxed{42 \text{ weeks}}$$

6.3 Activity Schedule

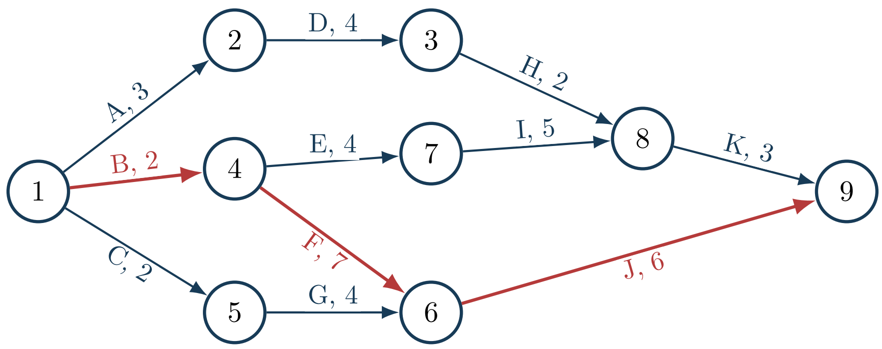
Activity	Predecessor(s)	Duration	ES	EF	LS	LF	TF
A	–	5	0	5	0	5	0
B	A	2	5	7	5	7	0
C	A	6	5	11	23	29	18
D	B	12	7	19	7	19	0
E	D	10	19	29	19	29	0
F	D	9	19	28	21	30	2
G	D	5	19	24	22	27	3
H	B	9	7	16	18	27	11
I	C, E	1	29	30	29	30	0
J	G	2	24	26	28	30	4
K	F, I, J	3	30	33	30	33	0
L	K	9	33	42	33	42	0
M	H, G	7	24	31	27	34	3
N	M	8	31	39	34	42	3

Change in duration

If activity G increases from 5 weeks to 10 weeks, the project duration becomes **44 weeks**. The new critical path is **A-B-D-G-M-N**. This demonstrates that the critical path can change when activity durations change.

7 Worked CPM Example 2: AOA Network and Floats

The following Activity-on-Arrow network is used to determine the critical path and the three principal floats.



The path durations are:

$$\begin{aligned}1-2-3-8-9 &: 3+4+2+3=12,\\ 1-4-7-8-9 &: 2+4+5+3=14,\\ 1-4-6-9 &: 2+7+6=\boxed{15},\\ 1-5-6-9 &: 2+4+6=12.\end{aligned}$$

Therefore, the critical path is **1-4-6-9**, corresponding to activities **B-F-J**, and the project duration is **15 days**.

Activity	Dur.	ES	EF	LS	LF	TF	FF	Raw IF	IF used
A	3	0	3	3	6	3	0	0	0
B	2	0	2	0	2	0	0	0	0
C	2	0	2	3	5	3	0	0	0
D	4	3	7	6	10	3	0	-3	0
E	4	2	6	3	7	1	0	0	0
F	7	2	9	2	9	0	0	0	0
G	4	2	6	5	9	3	3	0	0
H	2	7	9	10	12	3	2	-1	0
I	5	6	11	7	12	1	0	-1	0
J	6	9	15	9	15	0	0	0	0
K	3	11	14	12	15	1	1	0	0

8 Advantages and Limitations of CPM

8.1 Advantages

1. Especially useful for scheduling and controlling large projects.
2. Conceptually simple and easy to understand.
3. Graphical networks highlight relationships among project activities.

4. Identifies critical activities.
5. Applicable to many types of projects.
6. Helps monitor project time and cost.

8.2 Limitations

1. Activities must be clearly defined.
2. Precedence relationships must be known.
3. Time estimates may be subjective.
4. Excessive attention to the critical path may cause noncritical activities and resource interactions to be overlooked.

9 PERT and Variability in Activity Time

Activity time is not always fixed. For example, excavation might take 4 days under ideal conditions, 5 days under normal conditions, or 8 days during heavy rain. PERT represents this uncertainty by using three estimates and a beta-distribution approximation.

9.1 Three Time Estimates

Optimistic time, a Minimum reasonable duration if everything goes very well and no serious delay occurs.

Most likely time, m Most realistic duration under normal conditions; the mode of the assumed beta distribution.

Pessimistic time, b Maximum reasonable duration if several difficulties occur, such as rain, material shortage or labour disruption.

9.2 Expected Time, Standard Deviation and Variance

PERT activity-time formulas

Expected activity time:

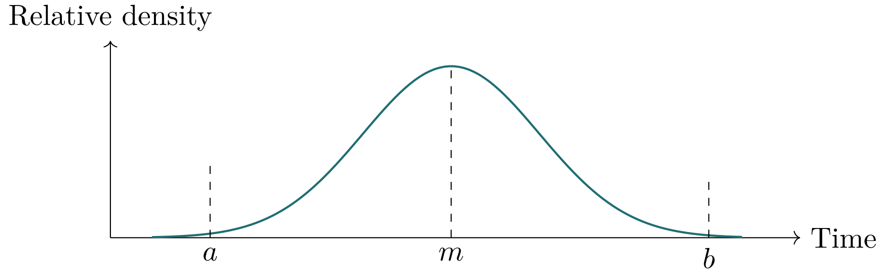
$$t_e = \frac{a + 4m + b}{6}$$

Standard deviation:

$$\sigma = \frac{b - a}{6}$$

Variance:

$$\sigma^2 = \left(\frac{b - a}{6}\right)^2$$



10 Project-Level PERT Calculations

After calculating the expected duration of every activity, determine the critical path using the expected times.

Project expected time, variance and standard deviation

If the critical path contains activities $1, 2, \ldots, n$:

$$TE = \sum_{k=1}^{n} t_{e,k}$$

Assuming activity durations are independent:

$$\sigma_p^2 = \sum_{k=1}^{n} \sigma_k^2$$

$$\sigma_p = \sqrt{\sigma_p^2}$$

By the normal approximation, the probability of meeting a target date T is found from:

Probability calculation

$$Z = \frac{T - TE}{\sigma_p}$$

where T is the target or due date, TE is expected project completion time, and σ_p is project standard deviation.

Useful standard-normal values:

Probability	50%	90%	95%	99%
Z	0.000	1.282	1.645	2.326

For a required probability p, the corresponding target duration is:

$$T = TE + Z_p \sigma_p.$$

11 Ten Steps of PERT

1. List all project activities and their immediate predecessors.
2. Draw a rough PERT network, identifying predecessors, successors and concurrent activities.
3. Draw the final network according to network rules.
4. Number events from left to right when using an AOA representation.
5. Estimate the three times a, m and b for every activity.
6. Calculate expected activity time $t_e = (a + 4m + b)/6$.
7. Use expected times as activity durations and calculate ES and EF by a forward pass.
8. Calculate LS, LF and floats by a backward pass; identify the critical path.
9. Calculate the variance of every activity.
10. Calculate project variance, project standard deviation and the probability of meeting the target completion date.

12 Difference Between CPM and PERT

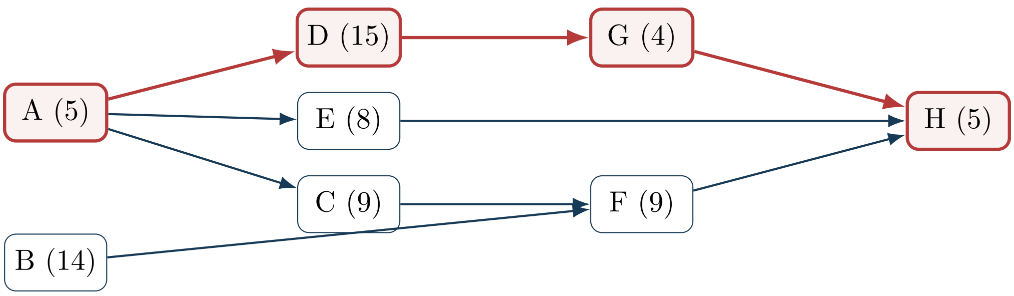
Feature	CPM	PERT
Nature	Deterministic	Probabilistic
Activity time	One fixed time estimate	Three estimates: optimistic, most likely and pessimistic
Time estimate used	Single duration	Expected duration calculated from a, m, b
Main focus	Time and cost optimization	Time estimation and probability of completion
Cost analysis	Commonly emphasized and may support crashing	Usually not the principal emphasis
Probability analysis	Not normally available from the basic method	Available using variance, standard deviation and the normal approximation
Critical path	Fixed for a given set of durations; changes if durations change	Based on expected times and may change as estimates are revised
Data required	Less	More
Complexity	Simpler	More complex
Typical objective	Complete within minimum feasible time and cost	Estimate project duration under uncertainty and assess deadline probability
Typical application	Construction, production and routine projects	Research, development and one-time uncertain projects

13 Worked PERT Example

13.1 Activity-Time Data

Activity	Predecessor(s)	a	m	b	t_e	Variance
A	–	2	4	12	5	2.78
B	–	10	12	26	14	7.11
C	A	8	9	10	9	0.11
D	A	10	15	20	15	2.78
E	A	7	7.5	11	8	0.44
F	B, C	9	9	9	9	0.00
G	D	3	3.5	7	4	0.44
H	E, F, G	5	5	5	5	0.00

13.2 Precedence Network



13.3 Expected-Time Schedule

Activity	t_e	ES	EF	LS	LF	TF	FF
A	5	0	5	0	5	0	0
B	14	0	14	1	15	1	0
C	9	5	14	6	15	1	0
D	15	5	20	5	20	0	0
E	8	5	13	16	24	11	11
F	9	14	23	15	24	1	1
G	4	20	24	20	24	0	0
H	5	24	29	24	29	0	0

Thus, the critical path is:

A-D-G-H

and the expected project duration is:

$$TE = 5 + 15 + 4 + 5 = \boxed{29 \text{ weeks}}.$$

13.4 Probability of Completion Within 30 Weeks

Only the variances of critical-path activities are added:

$$\sigma_p^2 = 2.78 + 2.78 + 0.44 + 0 = 6.00$$

$$\sigma_p = \sqrt{6} = 2.449 \text{ weeks}$$

For a target of $T = 30$ weeks:

$$Z = \frac{30 - 29}{\sqrt{6}} = 0.408$$

From the standard normal distribution:

$$P(\text{completion within 30 weeks}) \approx \boxed{0.659 \text{ or } 65.9\%}.$$

13.5 Duration Required for 99% Confidence

For 99% confidence, $Z = 2.326$:

$$T = 29 + 2.326\sqrt{6} = 34.70 \text{ weeks}$$

Therefore, the project should be allowed approximately:

$$\boxed{35 \text{ weeks}}$$

to have about a 99% probability of completion by the target date.

14 Quick Formula Sheet

CPM

$$EF = ES + d$$

$$ES = \max(EF_{\text{pred}})$$

$$LS = LF - d$$

$$LF = \min(LS_{\text{succ}})$$

$$TF = LS - ES = LF - EF$$

$$FF = \min(ES_{\text{succ}}) - EF$$

$$IF = \max\{0, E_j - L_i - t_{ij}\}$$

PERT

$$t_e = \frac{a + 4m + b}{6}$$

$$\sigma = \frac{b - a}{6}$$

$$\sigma^2 = \left(\frac{b - a}{6}\right)^2$$

$$TE = \sum t_e \text{ on CP}$$

$$\sigma_p^2 = \sum \sigma_i^2 \text{ on CP}$$

$$Z = \frac{T - TE}{\sigma_p}$$

$$T = TE + Z_p \sigma_p$$

Final exam reminders

- Always draw or verify the precedence network before calculating times.
- In a forward pass, use the **maximum** predecessor EF.
- In a backward pass, use the **minimum** successor LS.
- Critical activities have $TF = 0$.
- In PERT, sum variances - not standard deviations - along the critical path.
- A change in duration can change the critical path.

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